IMPT: A Multidimensional Transport Poverty Index

A replicable multi-dimensional and multi-modal measure of transport poverty

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Abstract

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1. Introduction

Transportation is an essential service which can condition access to almost all human activities, be them work, education, health or social life. When an individual or household has this access limited by any factor related to transport, be it insufficient supply, high costs or elevated exposure to externalities, the term Transport Poverty is used (Lucas

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et al., 2016). This phenomenon has direct consequences in social exclusion and territorial cohesion (Mejia-Dorantes and Murauskaite-Bull, 2022).

This problem has gained relevance over the last decade, particularly in the European context. The principle of access to essential services, including transportation, is enshrined in the European Pillar of Social Rights, and recently Regulation (EU) 2023/955, which established the Social Climate Fund (SCF) has provided the first legally bounding definition of Transport Poverty at the European scale, defining it as an "individuals' and households' inability or difficulty to meet the costs of private or public transport, or their lack of or limited access to transport needed for their access to essential socioeconomic services and activities, taking into account the national and spatial context". (Parliament and the Council of the European Union, 2023) .

Despite the clear empirical relevance of this phenomenon, there are difficulties in its systematic quantification. In Portugal, previous research has focused on punctual case studies (Pritchard et al., 2014) or on costly mobility surveys (INE, 2018), which allow neither for a continuous and longitudinal monitoring of this problem, nor its transferability to aid policy-makers' decisions.

This project was thus developed in an attempt to bridge the existing methodological gap with the creation of a composite and intermodal tool - the Multidimensional Transport Poverty Index (IMPT) - that is replicable and scalable, as well as wholly derived from public data, intended to assist and support public policy instruments, such as the European Union's Social Climate Fund (SCF) or Sustainable Urban Mobility Plans (SUMPs). The Lisbon Metropolitan Area (LMA) was employed as a case study for the tool's validation.

2. Literature review

2.1. Definitions of Transport Poverty

Transportation services are fundamental in modern society, and their deficiency or absence is related to social exclusion - a state where an individual or societal group is left out of participating in regular social activities in a given society (Pritchard et al., 2014) -, socio-territorial inequality and transport inequality - the differences in transport conditions for different groups of people (Hidayati et al., 2021). Despite the relation between poverty and transport conditions, these are also related to a wide range of social issues. As an example, motorized vehicle users contribute the most to externalities, but vulnerable road users are disproportionately affected by them in health, space allocation and time consumption (Gössling, 2016). Moreover, lower-income households tend to not have financial conditions to have motorized transport or to reside in central areas, relegating them to live in areas distant from the city center, often poorly served by public transportation. (Fedorowicz et al., 2020) (Lopes et al., 2023).

These differences in transport conditions can lead to transport poverty, a concept which lacks a single universally accepted definition in scientific literature. This expression has been used to broadly describe inequities related with transportation access, but also in a more restricted way to refer to the economic inability to support transportation costs (Mattioli et al., 2018). However, as Lucas et al. (2016) states, different terminologies relating to transport poverty - terms such as “mobility poverty”, “accessibility poverty” or “transport disadvantage” - are used with “often very different, although also overlapping definitions”. Due to this lack of a universal definition and the multidimensional nature of transport poverty, its measurement and quantification poses a challenge (Levinson and King, 2020).

This way, the necessity to comprehend this concept and quantify it becomes more urgent. It is essential to define reliable indicators of transport poverty in order to map disadvantaged areas, inform public interventions and monitor the implementation of measures to improve these situations. A reliable and replicable index to measure transport poverty is thus fundamental to support public policy instruments.

The most comprehensive conceptual systematization of transport poverty is proposed by Lucas et al. (2016), clearly defining previously different but overlapping terminology. They define transport poverty as a combination of four dimensions:

- Accessibility - the capacity to reach usual destinations and activities (such as work, education or health).
- Mobility - the systemic aspects of the transportation system itself that impact

essential trips, expressed in parameters such as public transport service frequency or average commuting time.

- Affordability - the ability of households to meet the cost of transportation.
- Exposure to externalities (Safety) - the negative effects of exposure to the transport system itself, such as environmental pollution or road crash rates.

Recently, and based on the aforementioned concepts, Mejia-Dorantes and Murauskaite-Bull (2022) have proposed a definition that has been widely adopted: transport poverty as the inability of an individual or household to access essential services or opportunities due to transportation services that are limited or non-existent, inadequate, unreliable, unaffordable, time-consuming or unsafe. This definition has been adapted and applied at the European scale, through Parliament and the Council of the European Union (2023) and a recent European Commission Report (European Commission. Directorate General for Employment and Inclusion, 2024). The key dimensions in the EC Report are synthesized as *Availability* (corresponding to Lucas' Mobility), *Accessibility* and *Affordability*, with exposure to externalities being included in the overarching concept of *Adequacy* (Figure 1).

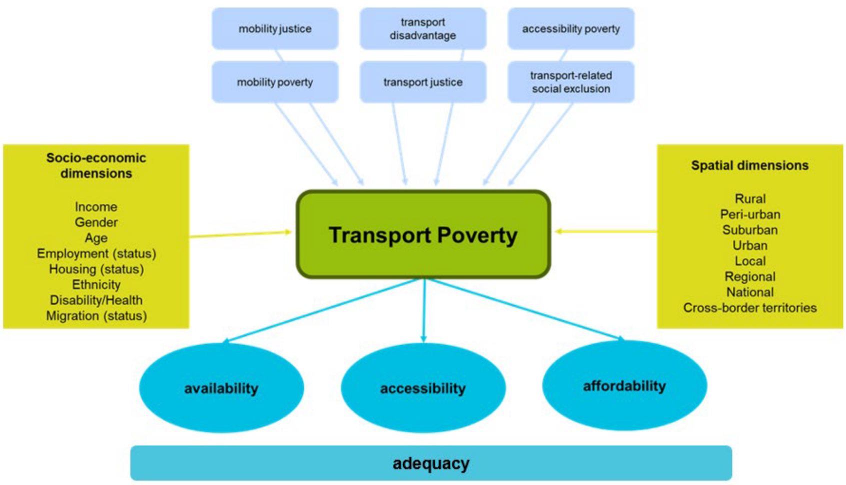


Figure 1: Conceptualization of Transport Poverty. Source: Oeko-Institut and University of Manchester

In summary, the concept of transport poverty has evolved from a one-dimensional look at accessibility or affordability, to a composite approach that attempts to fully capture the multidimensional nature of this phenomenon.

2.2. Dimensions of Transport Poverty

The dimensions proposed by Lucas et al. (2016) continue to be the reference for composite indexes of transport poverty (Alonso-Epelde et al., 2023) (Mejia-Dorantes and Murauskaite-Bull, 2022). IMPT adopts these dimensions, implementing them through secondary data available in Portugal. Each dimension is further explained thoroughly.

2.2.1. Accessibility

Accessibility refers to the ability to reach regular destinations such as employment, education, health and other essential services (Allen and Farber, 2019) (Unit, 2003). The literature distinguishes between approaches based on cumulative opportunities - the number of reachable destinations within a certain travel time threshold -, gravity-based approaches, which continuously weight destinations based on impedance (i.e. the easier a destination is to reach, the more weight it will have) and hybrid approaches (Geurs and van Wee, 2004) (Klar et al., 2023). Pereira and Herszenhut (2023) formalize these methodologies in open-source software *R*, through the library `r5r`, which provides the technical base for the Accessibility dimension of the IMPT (Pereira et al., 2021).

2.2.2. Mobility

The dimension of Mobility refers to the systemic characteristics of the transport system, be them existence and quality of infrastructure, service frequency or trip duration (Levinson and King, 2020) (Lucas et al., 2016) (Mejia-Dorantes and Murauskaite-Bull, 2022). This dimension captures different aspects of transport supply that accessibility is not able to, their core difference being that, while accessibility asks which places are reachable, mobility focuses on *how easy* it is to reach them (Levinson and King, 2020).

2.2.3. Affordability

This dimension determines the capacity of households to support essential transportation expenses (Isalou et al., 2014) (Litman, 2026) (Oviedo et al., 2025). Although different definitions of transportation affordability have been put forward (Lucas et al., 2016)

(Alonso-Epelde et al., 2023), IMPT uses the combined approach of *housing and transportation affordability*, based on the premise that households make their residential location choice based on both housing and transportation costs. If the combined housing and transportation cost doesn't exceed 45% of household income, it is considered affordable (Isalou et al., 2014) (Litman, 2026).

2.2.4. Safety

This dimension is commonly not included in transport poverty indexes (Kelly et al., 2023), being considered only in the most wide-ranging definitions of transport poverty (Lucas et al., 2016) (UN-Habitat, 2013). It refers to exposure to externalities of the transport system, such as road crashes and pollution, which have disproportionate effects on the territory and population. For the IMPT, this dimension's inputs only included those related to road safety indicators, due to the inability to get open and sufficiently granular data regarding environmental externalities.

A fifth dimension, digital literacy, was initially considered as a proxy for access to information and mobility services (Durand et al., 2022) (Orozco-Fontalvo et al., 2025) (Velaga et al., 2012). However, it was excluded from the final implementation of the index due to unavailability of data compatible with the scale of the lowest administrative division.

It is also worth noting that a consequence of IMPT's reliance on secondary data is the exclusion of another dimension, perceived accessibility, which the literature considers a robust indicator, but requires open and georeferenced data regarding individual and household preferences, which rarely exists (Farber), and is extremely difficult to monitor through time. This way, IMPT adopts an objective approach, computing all indicators and dimensions through openly available secondary data, sacrificing possible individual subjectivity in favor of replicability and simpler longitudinal monitoring.

2.3. Examples of geospatial tools of accessibility and equity

During the last decade various interactive spatial tools based on indicators of accessibility and inequality have been created and made publicly available. Two of these are here highlighted, due to their conceptual or methodological influence and their proximity to the IMPT's objective.

The project *Acesso a Oportunidades* (Access to Opportunities), developed by Instituto de Pesquisa Econômica (IPEA) under Rafael H. M. Pereira, provides accessibility estimates for the twenty biggest cities in Brazil, disaggregated in a hexagonal grid, by transport mode and type of opportunity. Its contribution is the creation of open R code, with the packages `r5r` and `accessibility`, which made the computation of multimodal travel time matrices possible (Pereira et al., 2021) (Pereira and Herszenhut, 2022) (Pereira and Herszenhut, 2023). An interactive tool to visualize these results was also created, and is available at: https://www.ipea.gov.br/acessooportunidades/mapa/. IMPT directly incorporates this technical infrastructure in its Accessibility dimension, in order to compute the number of reachable opportunities by different travel modes and travel time thresholds, for each grid cell in the study area. Pereira's work is this way a critial operational pillar of the IMPT.

Another example is the *Atlas of Inequality*, developed by MIT Media Lab and Northwestern University, under the coordination of Esteban Moro. This project is originated from a different problem, and attempts to measure *segregation by experience*, i.e. how people of different socioeconomic levels share the same urban spaces in daily life, through anonymized cell phone location data. This indicator thus captures inequality in the use of the city, and not transport supply or access to destinations. It is available at: https://inequality.media.mit.edu.

However, the methodological distinction between *Acesso a Oportunidades*, *Atlas of Inequality* and the IMPT is significant. *Acesso a Oportunidades* is a tool for the measurement of accessibility, i.e. what is reachable from each location, under assumptions on transport supply, but not taking into account the quality of the transport service offered, the economic capacity of households or exposure to externalities. The *Atlas of Inequality*, is a tool to measure segregation, which is often revealed by mobility patterns. It captures the social dimension of urban inequality without diagnosing transport poverty itself, due to the exclusion of other variables of transport supply, exposure to externalities and costs. IMPT, on the other hand, is a multidimensional index to measure transport poverty, aggregating the dimensions of Accessibility, Mobility, Affordability and Safety in a composite index, in line with the operational definition provided by the European Union (Parliament and the Council of the European Union, 2023) (European Commission. Directorate General for Employment and Inclusion, 2024) and supported by the literature (Lucas et al., 2016) (Mejia-Dorantes and Murauskaite-Bull, 2022).

2.4. IMPT's Contribution

IMPT emerges in an attempt to create an index synthesizing these dimensions and providing a reliable measurement of transport poverty, bridging three gaps in the literature and in national public policy.

1. There is, in Portugal, the absence of an index that integrates the four aforementioned dimensions across multiple transport modes and is replicable at the lowest administrative level.
2. Existing indexes excessively depend on costly primary data that is difficult to monitor continuously. IMPT relies exclusively on secondary and open data, easily allowing for updates and monitoring over time.
3. There is a lack of operational tools that allow to translate academic literature on transport poverty into decision-support instruments for local governments, transport authorities, and generally public policy makers.

Furthermore, the IMPT incorporates four methodological variants, corresponding to different analytical positions:

- **Contrasting IMPT:** based on entropy, it highlights the variables with the highest variation, better capturing territorial contrasts.
- **Critical IMPT:** based on the geometric mean, it non-linearly penalizes the dimension with the worst performance in each geographical unit.
- **Balanced IMPT:** based on the arithmetic mean, it allows for compensation between dimensions.
- **Political IMPT:** allowing for a personalized Analytic Hierarchical Process (AHP) weighting, it can be tailored to the decision-maker's preferences and priorities.

3. Methodology

Based on the guidelines provided by the OECD Handbook (OECD, 2008), the IMPT was computed through the following procedural steps: (1) data collection, computation and normalization of every indicator; (2) aggregation of indicators per dimension through a Principal Component Analysis (PCA), and normalization of each dimension; (3) multi-dimensional aggregation of the four dimensions, employing three distinct methods; (4) results validation, through a workshop with representatives of municipalities within the Lisbon Metropolitan Area (LMA).

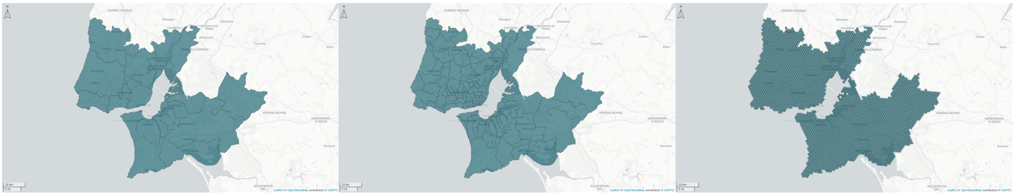
3.1. Case Study

The Lisbon Metropolitan Area (LMA) was applied as the case study for the IMPT. The LMA is comprised of two NUTS II regions (Grande Lisboa and Península de Setúbal), which encompasses 18 Municipalities and approximately 2,8 million inhabitants. This region presents territorial characteristics which make it particularly suitable for this analysis: a polycentric structure with a strong employment center in Lisbon, significant discrepancies between the public transport offer in the urban core and in the periphery, and a high modal share of private vehicles in a large part of the metropolitan area.

The analysis is focused on the lowest administrative division, the *freguesia* (parish), which is the most disaggregated geographical level that guarantees data coverage for all utilized sources. For the indicators that had a higher level of granularity, an analysis by grid level (see Figure 2c) was also assembled. The parish results were also aggregated by *município* (municipality), the administrative level immediately above the parish. The LMA contains 18 municipalities and 124 parishes (Figure 2), which constitute the observation units of the IMPT.

3.2. Data Sources

Based on the literature, a preliminary list of indicators for each dimension was compiled, followed by an extensive search for public data to obtain the most possible of these indicators. However, some of them were either not found or not publicly accessible. This way, the final list of indicators included is shown in Table 1. This excludes indicators such



(a) LMA Municipalities (b) LMA Parishes (c) LMA Grid

Figure 2: Geographical units used in the IMPT

as exposure to air pollution, overcrowding of public transportation, or walkability, as well as the dimension of digital literacy. The list of data sources used is shown in Table 2.

Table 1: Indicators used in the IMPT

Dimension	Indicator
Accessibility	Number of opportunities within travel time threshold
Accessibility	Travel time to first n opportunities
Mobility	Average commuting time
Mobility	Average number of commuting transfers
Mobility	Public transport headways and waiting times
Mobility	Public transport population coverage
Mobility	Ratio of pedestrian and cycling infrastructure
Mobility	Shared mobility availability
Affordability	Transportation costs
Affordability	Housing and transportation costs
Safety	Aggregation of road safety indicators

Table 2: Data sources used

Code	Database	Year	Source
CAOP	CAOP 2024.1 - Administrative Boundaries	2024	https://www.dgterritorio.gov.pt/sites/default/files/ficheiros-cartografia/CAOP_Continente_2024.1-gpkg.zip

Code	Database	Year	Source
BGRI21	Census 2021 - Base Reference Geographical Information	2021	https://mapas.ine.pt/download/filesGPG/2021/nuts3/BGRI21_170.zip
IMOB	IMOB 2017/18 - Inquérito à Mobilidade nas Áreas Metropolitanas	2018	https://www.ine.pt/xurl/pub/349495406
GTFS	GTFS — General Transit Feed Specification	2026-02	All LMA public transport operators (see Table 3)
OSM-RN	OpenStreetMap (OSM) — Road network	2026-01	https://export.hotosm.org/exports/4782f0b8-6778-4c0e-8e4f-97fc62e7f240
OSM-POI	OpenStreetMap (OSM) — Points of interest	2024-04	https://github.com/U-Shift/SiteSelection/releases/download/0.1/osm_poi_landuse.gpkg
CM	Carris Metropolitana — Health centres and schools datasets	2025	https://github.com/carrismetropolitana/datasets
ASNR	ANSR — Road accident data (location and type)	2019-2023	https://pmmus.tmlmobilidade.pt/
INE-Inc	INE — Household income by parish	2023	https://www.ine.pt/xurl/pub/66303599
INE-Hous	INE — Housing costs by parish	2021	https://www.ine.pt/xurl/pub/65586079
GBA	Global Building Atlas - Building heights	2026	https://github.com/zhu-xlab/GlobalBuildingAtlas
PM-MUS	TML PMMUS - Shared mobility docks	2025	https://pmmus.tmlmobilidade.pt/
COS	COS 2023 v1 — Land use map	2023	https://geo2.dgterritorio.gov.pt/cos/S2/COS2023/COS2023v1-S2-gpkg.zip

Code	Database	Year	Source
IP	Infraestruturas de Portugal — Toll costs	2026	https://portagens.infraestruturasdeportugal.pt

GTFS data sources for all LMA operators are presented in Table 3. This data includes public transport frequency, stop location and travel times, among other essential data to enable the estimation of Mobility and Accessibility metrics.

Table 3: GTFS sources for LMA public transport operators

Operator	Mode	GTFS URL
Carris Metropolitana (CM)	Bus	https://api.carrismetropolitana.pt/gtfs
Carris	Bus	https://gateway.carris.pt/gateway/gtfs/api/v2.8/GTFS
MobiCascais	Bus	https://drive.google.com/u/0/uc?id=13ucYiAJRtu-gXsLa02qKJrGOgDjbnUWX&export=download
Transportes Colectivos do Barreiro (TCB)	Bus	https://backend.tcbarreiro.pt/download-gtfs
Metropolitano de Lisboa (ML)	Metro	https://metrolisboa.pt/google_transit/googleTransit.zip
Comboios de Portugal (CP)	Suburban rail	https://publico.cp.pt/gtfs/gtfs.zip
Fertagus	Suburban rail	https://mts.pt/imt/MTS-20240129.zip
Metro Transportes do Sul (MTS)	Light rail	https://fertagus.pt/GTFSTMLzip/Fertagus_GTFS.zip
Transtejo/Softlusa (TTSL)	Ferry	https://api.transtejo.pt/files/GTFS.zip

Lastly, the following R packages were utilized in the data processing and creation of both the index and its interactive dashboard: `readr`, `readxl`, `openxlsx`, `lubridate`, `dplyr`, `tidyr`, `stringr`, `tidyverse`, `sf`, `terra`, `mapview`, `stplanr`, `h3jsr`, `r5r`, `accessibility`, `odjitter`, `tidytransit`, `GTFShift`, `FactoMineR`, `assertthat`, `piggyback`.

3.3. Each Dimension's Indicators

The IMPT is not only multidimensional, but also intermodal. In Table 4, the dimensions used to compute the index for each mode are specified.

Table 4: Modes computed for each dimension of the IMPT

Dimension	Mode
Accessibility	Driving, Public Transport, Walking, Cycling
Mobility	Driving, Public Transport, Walking, Cycling
Affordability	Driving, Public Transport
Safety	Driving, Walking, Cycling

3.3.1. Accessibility

The Accessibility dimension quantifies the capacity to reach key destinations within a given travel time threshold. The key destination categories, as well as their data sources, are summarized in Table 5.

Table 5: POI categories, sources and filters

Category	Identifier	Source	Filter/Description
Healthcare (all)	health	CM	All health centers
Hospitals	health_hospital	CM	Name contains 'Hospital' or 'Centro de Medicina'
Primary care centers	health_primary	CM	Name contains 'Centro de Saúde'

Category	Identifier	Source	Filter/Description
Supermarket/groceries	groceries	OSM	`type %in% c('supermarket','convenience')`
Green spaces	green_spaces	OSM	`group == 'leisure' & type %in% c('park','garden')`
Recreation	recreation	OSM	`type %in% c('library', 'theatre', 'cinema', 'restaurant', 'cafe', 'bar', 'pub', 'pitch', 'fitness_station', 'swimming_pool', 'sports_centre', 'fitness_center')`
Schools (all)	schools	CM	All school types
Primary schools	schools_primary	CM	Primary education schools only
Transit stops (all)	transit	GTFS (all operators)	All transit stops from all operators
Transit stops (bus)	transit_bus	GTFS (Carris, CM, MobiCascais, TCB)	Bus stops only
Transit stops (mass transit)	transit_mass	GTFS (ML, CP, Fertagus, MTS, TTSL)	Mass transit (metro, rail and ferry) stops only
Jobs	jobs	IMOB	Jittered OD destinations (see Section 3.3.1.1)

This dimension incorporates two metrics:

1. Cumulative number of opportunities: number of points of interest of a given category reachable within a certain amount of time.

2. Travel time to reach nearest opportunities: time required to reach the first n opportunities of a given category.

Indicators for this dimension were computed using the H3 grid (Uber Technologies, 2018) of resolution 8 (Pereira and Herszenhut, 2022), which corresponds to 3686 hexagons in the LMA, with an edge length of 531 meters and an average hexagon area of approximately 737,327m^2.

3.3.1.1. Jittering.

Due to a lack of open access to fine data regarding job location, the most granular data obtained was an OD matrix at the parish level, provided by INE (2018), with centroid to centroid representations, "a common approach involving the simplifying assumption that all trip destinations and origins can be represented by (sometimes population weighted or aggregated) zone centroids" (Lovelace et al., 2022). However, in order to accurately compute accessibility and mobility measures, there was the need to have more disaggregated data regarding job locations, beyond the parish unit.

This way, jittering (Lovelace et al., 2022) was used as a proxy for this purpose, using building volumes as trip attractors, and this way weighting origins (random samples from residential buildings) and destinations (buildings weighted by volume). This allowed for the disaggregation of trips between parishes to OD pairs with a maximum of 50 trips per OD, with ODs with more trips than this value are split into multiple point-to-point pairs, in order to provide further spatial resolution. The process generated approximately 18,500 jittered OD pairs that represent the daily commuting patterns of the LMA. This value was then expanded to the real number of trips using a weighting variable (*PESOFIN)* provided by INE (2018).

3.3.1.2. Travel time matrix.

A travel time matrix was used as the basis for all accessibility measures. It was computed between each grid cell, over a network (Pereira et al., 2021) that was built considering:

- OpenStreetMap (OpenStreetMap, 2026) road network, exported from HOT (2026).

- General Transit Feed Specification (GTFS) for all operators in the LMA (valid at 04/02/2026).

- Terrain elevation profile (EU, 2026).

The matrix was computed using the function `r5r::travel_time_matrix()` (Pereira and Herszenhut, 2023) with the parameters shown in Table 6.

Table 6: Parameters used in the computation of the travel time matrix

Modes	Parameter	Description
All	`percentiles = 50`	50th percentile (median) travel time. It is used so results represent the median travel conditions and not the "best-case" or "worst-case" scenarios.
All	`max_trip_duration = 120`	Maximum trip duration, in minutes.
Public transport	`mode_egress=WALK`	Mode used after egress from the last public transport.
Public transport	`max_walk_time = 15`	Maximum walking time (in minutes) to access or egress public transport, or to transfer inside the network.
Public transport	`departure_datetime = 04/02/2026 at 08:00:00 AM`	Departure date and time considered for public transport.
Public transport	`Max_rides = 2` (1 transfer) or `Max_rides = 3` (2 transfers)	Maximum number of public transport transfers allowed in the same trip.
Cycling	`max_lts = 3`	Maximum level of stress tolerated by cyclists in a road environment.

3.3.1.3. Cumulative number of opportunities.

The cumulative number of opportunities was calculated using the function `accessibility::cumulative_cutoff()` considering the travel time matrices for each grid cell combination, the number of opportunities per grid cell, and the travel time cut-offs.

This measure was computed for all combinations of POIs and modes for time cut-offs of 5, 10, 15, 30, 45, 60, 75 and 90 minutes. However, as can be seen in Table 7, only selected combinations of modes/points of interest were included in IMPT's operationalization.

Table 7: Travel time threshold (minutes) to compute cumulative number of opportunities

Point of Interest	Driving or Public Transport	Walking or Cycling
Healthcare facilities	30	15
Basic education	30	15
Green spaces	5-10	15-30
Recreation	5-10	15-30
Groceries	15	15
Jobs	30-45	30-45

The value computed represents the number of opportunities reachable within the defined travel time threshold. Values at parish and municipality level are aggregated considering a weighted mean of the number of opportunities reachable in each grid cell within that geographical unit, weighted by the population of those cells.

3.3.1.4. Travel time to reach closest opportunities.

This measure was computed using the function `accessibility::cost_to_closest()` (Pereira and Herszenhut, 2022) , considering the travel time matrices for each grid cell combination and the number of opportunities in each grid cell. Its output corresponds to the travel cost, in minutes, to reach n opportunities of a certain category. Although calculations were performed for 1, 2 and 3 opportunities, only some of these were included in IMPT's operationalization, depending on the POI category:

- Healthcare, basic education and green spaces: 1 opportunity.

- Recreation, groceries and jobs: 3 opportunities.

3.3.2. Mobility

Mobility evaluates the quality of the transportation system as a service, independently of the accessibility of the destinations. The methodology to calculate its principal indicators is explained and detailed below.

3.3.2.1. Average commuting time.

Using the jittered (see Section 3.3.1.1) OD pairs from INE (2018) (for commuting) and the travel time matrices, average commuting travel times are computed for each geographical unit and mode, weighted by number of trips. Trips with no feasible route within the constraints are assigned the maximum travel time (120 min) and counted separately as "not accomplishable" trips (PNA - Percentagem Não Alcançáveis).

3.3.2.2. Average number of commuting transfers.

For each commuting OD pair, the minimum number of transfers required for each public transport route (for 0, 1, 2, and 3 transfers) is assigned to each OD pair, and population-weighted mean transfers are computed per geographical unit. The result is the average minimum amount of transfers to commute to work using public transport.

3.3.2.3. Public transport headways and waiting times.

Stop-level service frequencies were computed using `tidytransit::get_stop_frequency()` (Poletti et al., 2026) for four different time windows, as can be seen in Table 8.

Table 8: Public transport service frequency periods observed

Period	Day	Time window
Weekday peak	Wednesday 04/02/2026	08:00-09:00
Weekday full day	Wednesday 04/02/2026	00:00-48:00
Weekday night	Wednesday 04/02/2026	22:00-23:00
Weekend	Sunday 08/02/2026	10:00-11:00

A 500-meter buffer per stop is used to allocate stops to BGRI (INE, 2021a) population centroids. Results are aggregated to geographical units as population-weighted mean headway and waiting time. Additionally, frequency reduction factors were computed to quantify the reduction in service during night and weekend periods.

3.3.2.4. Public transport population coverage.

This indicator is estimated using walking isochrones around public transport stops (see Figure 3a as an example), differentiated by transport mode and their typical service area. Isochrones are computed with the function `r5r::isochrone()` using the `WALK` mode and the AML street network. Three mode groups were defined based on stop type and catchment area, the rational being that high-density networks (e.g. buses) have a lower acceptable walking threshold, while mass transit networks with sparse stops (e.g. suburban railways) induce higher walking thresholds:

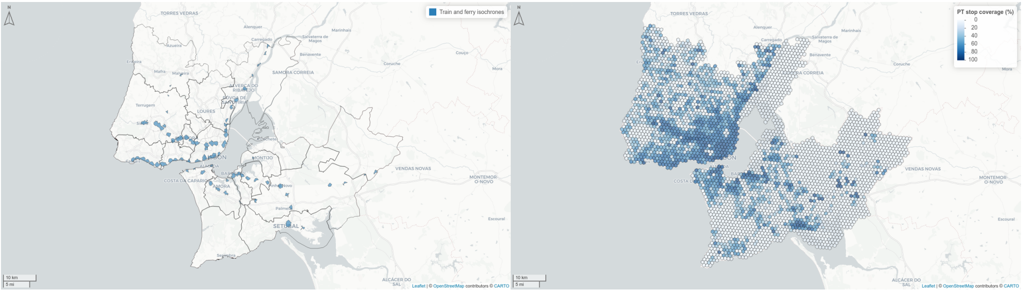
- Bus (Carris, CM, MobiCascais, TCB) - 5-minute walking threshold.
- Metro & light rail (ML, MTS) - 10-minute walking threshold.
- Train & Ferry (CP, Fertagus, TTSL) - 15-minute walking threshold.

To calculate resident population beyond the parish level, BGRI21-level population data (INE, 2021a) was further disaggregated by residential density class through the intersection with COS (DGT, 2023), in order to have the highest-possible resolution for residential locations. The residential density classes considered are shown in Table 9.

Table 9: Population and building density weights for COS land use classes

Class of residential area	Translation	Weight
Contínuas predominantemente verticais	Continuous and predominantly vertical	3.0
Contínuas predominantemente horizontais	Continuous and predominantly horizontal	1.0
Descontínuas	Discontinuous	0.6
Descontínuas esparsas	Discontinuous and sparse	0.3

Population data was then intersected with the public transport stop isochrones, and aggregated by geographical unit (see Figure 3b).



(a) Walking isochrones to train and ferry stations   (b) Total coverage of public transport stops

3.3.2.5. Ratio of pedestrian and cycling infrastructure.

The pedestrian infrastructure ratio is the proportion of road length that includes pedestrian infrastructure, computed using OpenStreetMap (OpenStreetMap, 2026) data:

$$R_{ped} = \frac{pedestrian\ path\ length}{road\ length}$$

Road classification uses the highway tags: `motorway, trunk, primary, secondary, tertiary, unclassified, residential, living_street`, as well as linked variants.

Pedestrian infrastructure classification uses the tags: `highway = footway | pedestrian | steps | residential` and `footway = sidewalk | crossing | path | platform | corridor | alley | track, and sidewalk = both | left | right`.

The calculation of cycling infrastructure ratio is analogous to the pedestrian ratio, and also extracted from OpenStreetMap (2026). Cycling infrastructure is further classified according to its type and level of segregation, according to the following description and OSM tags:

- Cycle track or lane (*Ciclovia segregada*): Segregated or light-separated tracks exclusive for cycling.

  - `highway = cycleway`
  - `highway = path` & `bicycle = designated`
  - `segregated = yes | no`
  - `cycleway` tags containing: `separate`, `buffered_lane`, `segregated`, `lane`, or `track`.

- Advisory lane (*Via partilhada com veículos motorizados / zonas 30*): Marked cycle lanes shared with motor vehicles, such as "sharrows" or advisory lanes.
  - `cycleway` tags containing: `shared_lane`.

- Protected Active (*Via partilhada com peões*): Infrastructure shared with pedestrians but protected from motor vehicles.
  - `highway = pedestrian` & `bicycle = designated`.
  - Refinement of `highway = path | footway | pedestrian` when `foot` access is explicitly allowed on cycling paths without physical segregation.

The primary indicators are thus:

- **Cycling infrastructure ratio:**

$$R_{cyc} = \frac{total\ cycleway\ length}{road\ length}$$

- **Cycling quality ratio:**

$$R_{ped} = \frac{segregated\ cycleway\ length}{total\ cycleway\ length}$$

Where "segregated cycleway length" corresponds to the "cycle track or lane" category.

3.3.2.6. Shared mobility availability.

The number of shared mobility points (physical and virtual docks from all operators in the LMA (TML and Way2Go, 2025)) was counted and aggregated per parish and per grid cell.

3.3.3. Affordability

Affordability evaluates the capacity of households to support the monetary costs of transportation to access essential activities. IMPT uses a weighted cost by mode, in which the monetary cost of automobile/public transport travel is multiplied by each parish's modal share, and expressed as a proportion of median household income. The travel costs are estimated for one-way trips from home to the workplace during the morning peak.

3.3.3.1. Cost of driving.

The cost of car travel is computed through:

$$Cost_{car} = d \times 0.40€/km + Cost_{toll}$$

Where a fixed cost of 0.40€/km is used, based on República Portuguesa (2008), which sets mileage allowance to cover fuel and car maintenance. Tolls are computed through the toll calculator of Infraestruturas de Portugal (IP, 2026), using the route geometry returned by `r5r::detailed_itineraries()`.

3.3.3.2. Cost of public transportation.

To account for uncertainty in public transport fare payments, two scenarios were considered:

Scenario A - Single ticket

The average single ticket cost was calculated based on revenue data from Carris (Carris, 2024):

- Total single ticket validations: 17,424,000

- Total single ticket revenue: 33,278,000€
- Average cost per single ticket: ~1.91€

This way, a single ticket cost of **1.91€** was assumed for all modes. Free transfers within the same mode for 60 minutes are assumed, but transfers to different modes incur an additional charge.

Scenario B - Monthly pass (Navegante)

Based on operator revenue data, the average cost of a monthly pass was estimated (Table 10).

Table 10: Revenue from Navegante monthly pass

Pass type	Pass recharges	Pass price
Navegante Metropolitano	5,063,510	40€
Navegante Municipal	887,190	30€
Senior (+65)	1,103,782	20€
Family	243,784	80€
Youth (sub-23)	3,425,013	0€

The average cost paid by a monthly pass holder is thus ~**25.25€**. Assuming an average of 44 monthly trips (2 trips in each of 22 working days), the average cost per trip becomes ~**0.57€**.

3.3.3.3. Composite indicators.

Transport expenditure is expressed as a share of household income, with two variants (with and without housing costs). All costs are annualized and computed as yearly expenses per household (€/year/household). 250 days per year were considered, with two trips per day. Tolls are considered assymetrically for both directions, since the bridges' tolls are only paid one-way.

For driving:

$$transp\_inc_{car} = \frac{C_{car}}{I_{household}}$$

$$h\_transp\_inc_{car} = \frac{C_{car} + C_{housing}}{I_{household}}$$

To combine car and public transport costs, a composite indicator based on each parish's modal share from INE (2021b) is computed:

$$transp\_inc_{weighted} = \frac{C_{car} \cdot s_{car} + C_{PT} \cdot s_{PT}}{s_{car} + s_{PT} + s_{active}}$$

Where s_m is the modal share of mode m. Active modes (walking and cycling) are included in the denominator to preserve the real modal structure of each parish, but are not included in the numerator, given their null direct cost. This option avoids overestimating the financial burden in parishes with a high active modal share, while reflecting the fact that a part of the population travels without a cost.

Driving costs are further adjusted for vehicle occupancy rate, household size and mobile population fraction (INE, 2018), to convert per-trip costs to a household-level annual burden. This way, the final average annual transport expenditure, accounting for all adjustments and modal shares, is computed as:

$$transp\_inc\_comp = \frac{HH_{size} \cdot P_{mobile}}{I_{household}} \cdot \left[\left(\frac{C_{car,day}}{O} \cdot \frac{s_{car}}{s_{car} + s_{PT} + s_{active}} \right) + (C_{PT,day} \cdot \frac{s_{PT}}{s_{car} + s_{PT} + s_{active}}) \right] \cdot N_{days}$$

Where:

- $N_{days} = 250$ working days per year.
- HH_{size} is the average household size.
- P_{mobile} is the share of mobile population.
- $I_{household}$ is the average annual household income.

- $C_{car,day}$ and $C_{PT,day}$ are the round-trip costs for car and public transport.

- O is the average vehicle occupancy rate.

- s_{car}, s_{PT} and s_{active} are the modal shares (INE, 2021b).

For the variant that includes housing costs ($h\_transp\_inc\_comp$), the annualized housing cost $C_{\text{housing,year}}$(INE, 2022) is added to the numerator.

3.3.4. Safety

This dimension encompasses the disproportional exposure to externalities of the transportation system. It is entirely based on road safety data provided by Autoridade Nacional de Segurança Rodoviária (ASNR) through Transportes Metropolitanos de Lisboa (TML) (TML and Way2Go, 2025). It is important to note that only road crashes occuring *inside localities* (dentro das localidades), a category specified by the ASNR itself, are included in the index. In particular, this excludes crashes on motorways and on the two bridges crossing the Tagus river, while maintaining IMPT's spatial scope.

The indicators calculated for each geographical unit and transport mode are systematized below:

- **Total accidents (5 years):**
$$\sum(Acc_{5y})$$
- **Fatalities (30-day):**
$$\sum(VM_{30d})$$
- **Serious injuries (30-day):**
$$\sum(FG_{30d})$$
- **Slight injuries (30-day):**
$$\sum(FL_{30d})$$
- **Accident rate:**
$$\frac{\sum(Acc_{5y})}{Population} \times 1000$$
- **Fatality rate:**
$$\frac{\sum(VM_{30d})}{Population} \times 1000$$

- **Severity index (total):**

$$\frac{\sum(VM_{30d})}{\sum(Acc_{5y})}$$

- **Severity index (motorized):**

$$\frac{\sum(VM_{30d})}{Motorized\ vehicles\ involved}$$

- **Severity index (cycling):**

$$\frac{\sum(VM_{30d})}{Cyclists\ involved}$$

- **Severity index (pedestrian):**

$$\frac{\sum(VM_{30d})}{Pedestrians\ involved}$$

3.4. Normalization and Aggregation

All indicators were normalized to the range [1, 100] using min-max scaling:

For benefit indicators (higher raw value → better):

$$X_{norm} = 1 + \frac{X - X_{min}}{X_{max} - X_{min}} \times 99$$

For cost indicators (lower raw value → better):

$$X_{norm} = 1 + \frac{X_{max} - X}{X_{max} - X_{min}} \times 99$$

- **Accessibility** metrics (`access_*`) are *benefit* indicators.
- **Mobility** cost metrics (`mobility_cost_*`), public transport waiting times, all **Affordability** indicators, and all **Safety** indicators are *cost* indicators.

Missing values in accessibility indicators (arising from unfeasible public transportation routes) are imputed with the mean of the respective variable prior to PCA.

3.5. PCA-Based Dimension Scores

For every dimension with multiple indicators (Accessibility, Mobility and Safety) and at the global (all-mode) and per-mode levels, a PCA was run using `FactoMineR::PCA()` at parish level.

In line with OECD (2008), the first principal component (PC1) score is used as the dimension index after rescaling to [1, 100]. PC1 captures the dominant axis of variation among the normalised indicators.

For the Affordability dimension, a PCA is not applied, due to there being too few indicators. Instead, the normalized composite `h_transp_inc_comp` (housing + transport cost as share of income, modal-share weighted) is used directly as the Affordability Index.

For Safety per mode, an equal-weight average of the severity index (per mode) and total vehicles involved (per mode) is used instead of PCA.

For Mobility - Car, a PCA is not applied, also due to there being too few indicators. Instead, we assumed an equal-weight average of the normalized composite `mobility_commuting_avg_tt_car` and `road_length`.

3.6. Composite IMPT Scores

Three aggregation methods are computed for the global IMPT and each per-mode IMPT:

1. **Critical IMPT - geometric mean** (equal weights, Figure 4a):

$$IMPT_{geom} = \left(\prod_{d} (S_d + 1) \right)^{1/n} - 1$$

Partially compensatory: penalizes dimensions with very low values non-linearly. Highlights the most critical situations, where at least one dimension is severely deficient.

2. **Balanced IMPT - arithmetic mean** (equal weights, Figure 4b):

$$IMPT_{avg} = \frac{1}{n} \sum_{d} S_d$$

Compensatory: high vulnerability in one dimension can be offset by good performance in others. Suitable for setting aggregate policy targets and communicating to non-specialist audiences.

3. **Contrasting IMPT - entropy-weighted geometric mean** (Figure 4c):

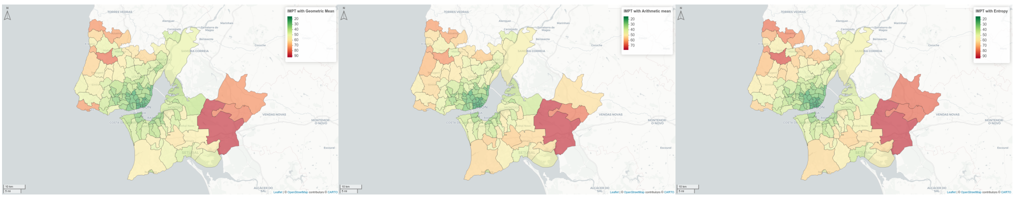
$$IMPT_{entropy} = \prod_{d} (S_d + 1)^{w_d} - 1$$

where weights w_d are derived from the **entropy weighting method**:

Weights are determined by the empirical dispersion of the dimensions across parishes: the dimension that varies most among territories receives greater weight. Maximizes territorial contrast and is useful for competitive prioritization of interventions.

$$w_d = \frac{d_d}{\sum_j d_j}, \quad d_d = 1 - e_d, \quad e_d = -k \sum_i p_{id} \ln(p_{id})$$

with $k = 1/\ln(n)$ and p_{id} being the proportion of the dimension score for area i.



(a) Critical IMPT (b) Balanced IMPT (c) Contrasting IMPT

Figure 4: IMPT results for different aggregation methods for all modes (with Navegante)

3.7. Validation

Finally, a workshop was carried out with LMA municipalities. It took place on April 9th, 2026 in the offices of Transportes Metropolitanos de Lisboa (TML), with the objectives of not only presenting and validating the results with municipality technicians, but also confronting the computed data with the local perception of the participants, in order to identify relevant divergences. It included representatives from 13 out of the 18 municipaities in the LMA, as well as representatives from transport authorities and operators.

4. Results

The full results of the IMPT for all LMA parishes are available in the dashboard (Figure 5) created in the scope of this project. This dashboard allows for the visualization of the global indices (critical, balanced and contrasting), the results of each dimension by transport mode, all indicators that make up the index, as well as key statistics and summaries for various scales (parish, municipality and grid). Beyond results visualization, this tool allows for the calculation of the aggregated index adjusted to user preferences (political IMPT). For this purpose, an Analytic Hierarchical Process (AHP), that compares the user's preferences between dimensions and this way assigns weights to each, is made available to the dashboard's users. The dashboard is available at: https://ushift.tecnico.ulisboa.pt/apps/impt.

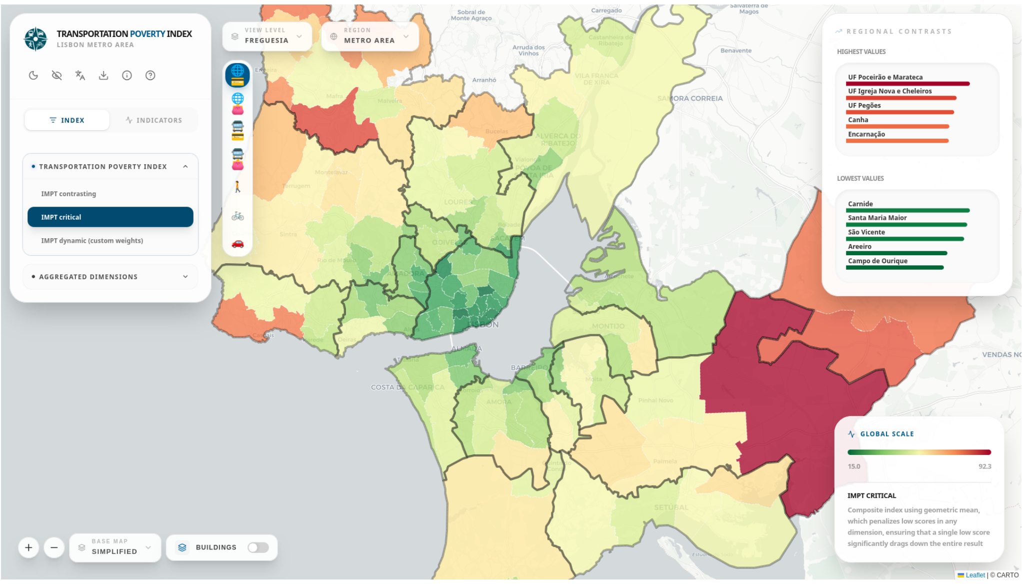


Figure 5: IMPT Dashboard

It is worth noting that the IMPT is a relative index, created with the purpose of comparing different areas in the same metropolitan area, not of being an absolute index of transport poverty. This way, a global score of 0 (resp. 100) should not be interpreted as the "best" (resp. "worst") possible outcome.

4.1. Methodological comparison

From Table 11, it can be seen that the five parishes with the lowest contrasting IMPT form a contiguous cluster in the center of Lisbon. These areas benefit simultaneously from high Accessibility and Mobility (being located in the urban core, they have very good access to opportunities and quality public transport service and active modes infrastructure) and relatively low road accident rates (high Safety). Although these parishes also score very high in the Affordability dimension, it is worth noting that this score also stems from the high median income in this region, and is not only due to low transportation costs.

Meanwhile, the parishes which score the highest in the IMPT are predominantly located in rural regions in the east and northwest, far from the urban core, namely in the municipalities of Montijo and Mafra. These parishes tend to score very poorly in every dimension apart from Safety, which is oftentimes even higher than the LMA average. The concentration of the worst-scoring parishes in specific areas of the LMA suggest that integrated intervention strategies, at the municipal level or higher, can be more effective than parish-by-parish interventions.

Table 11: Contrasting IMPT results for the five best and worst-ranked parishes in the LMA

Ranking	Parish	Contrasting IMPT
1	Arroios	13.4
2	Areeiro	14.9
3	Campo de Ourique	15.0
4	Santo António	15.3
5	Santa Maria Maior	15.6
...	...	...
120	Encarnação	81.8
121	Canha	83.8
122	União de Freguesias de Pegões	83.9
123	União de Freguesias de Igreja Nova e Cheleiros	90.0
124	União de Freguesias de Poceirão e Marateca	93.4

While the contrasting IMPT is a good indicator to evaluate differences between territories, the critical IMPT is better for evaluating specific dimensional deficiencies in each geographical unit.

Analyzing this index (see Table 12), it is visible that Arroios, which lead the contrasting IMPT ranking, is now outside of the five lowest-scoring parishes due to a poorer result in the Safety dimension, which was previously offset by a very high score in the other three. Carnide, also located close to the urban core of Lisbon, surges from 11th place in the contrasting IMPT to 5th, benefiting from having consistent scores in all four dimensions. Meanwhile, the highest-scoring parishes remain in the same rankings, with the only difference being the existence of slightly higher variation between their scores.

Table 12: Critical IMPT results for the five best and worst-ranked parishes in the LMA

Ranking	Parish	Critical IMPT
1	Campo de Ourique	15.0
2	Areeiro	15.5
3	São Vicente	18.1
4	Santa Maria Maior	18.5
5	Carnide	18.9
...	...	...
120	Encarnação	76.6
121	Canha	77.0
122	União de Freguesias de Pegões	80.6
123	União de Freguesias de Igreja Nova e Cheleiros	82.4
124	União de Freguesias de Poceirão e Marateca	92.3

4.2. Dimensional and spatial performance

4.2.1. Accessibility

The main deficiencies in Accessibility in the LMA are located in rural parishes, far from any urban center, where reaching essential services is often a challenge due to low density and long distances to essential services. Meanwhile, most urban centers perform well on this dimension, given that all urban areas provide a significant concentration of essential services, which are usually accessible in a reasonable travel time.

4.2.2. Mobility

This dimension indicates areas where either there is a lack of transport alternatives, or even if they exist, they are systemically inadequate, having impairments such as low service frequency or high travel times. The results for this dimension highlight the radial nature of the Lisbon Metropolitan Area, with travel from the urban core of Lisbon (where the highest concentration of jobs exists) being unchallenging, but with travel from peripheral regions being much more constrained.

4.2.3. Affordability

Most parishes of the LMA perform well in this dimension, showing high values of transport affordability. The most critical cases are concentrated in lower-income, car-dependent areas. This way, it can be stated that transport affordability isn't the primary concern for the LMA. The positive scores in this dimension can be at least partially attributed to the existence of a simple monthly, multimodal and wide-ranging public transport pass, the Navegante, which contributes significantly to lowering household transportation costs.

4.2.4. Safety

This dimension exhibits different territorial patterns from the others, with peripheral parishes performing mostly better than centrally-located ones. The poor results of some parishes in the Lisbon core is a result of the high number of crashes and vehicles involved, even if their severity is not extreme. However, the worst results are displayed in the near-shore areas of Oeiras and Cascais, and can be attributed to the particularly high crash and mortality rates registered in the Marginal road (EN6).

4.3. Takeaways

5. Discussion

5.1. Limitations

6. Conclusions

This is the best paper ever.

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